Placement Preparation Roadmap by Ajays_vision_

Hey there 👋

If you're getting ready for placements, this roadmap will help you cover everything step by step — from aptitude to HR.

Let's make it simple and focused.

🧮1. Aptitude Round

Main Goal: Get fast and accurate with numbers — because this is the first filter in almost every company.

Best Resource: [IndiaBix]

Topics to Focus On:

- Numbers, Simplification & Approximation
- Percentages, Ratio & Proportion, Averages
- Simple & Compound Interest, Profit & Loss
- Time, Speed & Distance, Time & Work
- Pipes & Cisterns, Boats & Streams
- Probability, Permutation & Combination
- Algebra, Geometry, Mensuration
- Data Interpretation
- Logical Reasoning → Puzzles, Seating Arrangement, Blood Relations, Coding-Decoding, Syllogisms

Tip:

Start with 2–3 topics a week. Focus on understanding first, then speed.

Take short mock tests on weekends to check your progress.

2. Technical Core Subjects

This round tests how well you understand your basics — not how much you've memorized.

So focus on concept clarity and real-life examples.

Resource: GeeksforGeeks (you'll find everything here!)

⚙️ Operating System

- Process vs Thread
- Deadlock (conditions, prevention, detection)
- Scheduling Algorithms – FCFS, SJF, Round Robin, Priority
- Paging, Segmentation, Virtual Memory
- Page Replacement Algorithms
- Inter-Process Communication (IPC)

🌐 Computer Networks

- OSI vs TCP/IP
- TCP vs UDP
- IP Addressing (IPv4, IPv6, Subnetting basics)
- DNS, DHCP, ARP
- Congestion Control
- HTTP vs HTTPS
- Hub vs Switch vs Router

🗄️ DBMS

- Normalization (1NF–BCNF)
- SQL Queries – Joins, Subqueries, Group By, Having
- Transactions (ACID properties)
- Primary Key, Foreign Key, Unique Key
- Triggers, Views, Stored Procedures
- ER Model basics

🧠 **OOPS Concepts**

- 4 Pillars: Encapsulation, Abstraction, Inheritance, Polymorphism
- Overloading vs Overriding
- Abstract Class vs Interface
- Constructor vs Destructor
- Real-life examples
- Access Specifiers

Pro Tip: Try to relate every concept to a real-world example

it helps you remember and explain better in interviews.

3. Data Structures & Algorithms (DSA)

This is where you stand out.

Companies love candidates who can think logically and code efficiently.

Resource: LeetCode (or CodeStudio if you prefer topic-wise problems)

Must-Know Topics:

- Arrays: Two Pointers, Sliding Window, Prefix Sum

- Strings: Palindrome, Anagram, Pattern Matching
- Hashing: Frequency Count, Subarray Sum
- Recursion & Backtracking: Subsets, Permutations, N-Queens
- Searching & Sorting: Binary Search, Merge Sort, Quick Sort
- Linked List: Reversal, Cycle Detection, Merge Two Lists
- Stack & Queue: Next Greater Element, Min Stack, LRU Cache
- Binary Tree/BST: Traversals, LCA, Diameter
- Heap: Kth Largest/Smallest, Top-K Elements
- Graphs: BFS, DFS, Dijkstra
- Dynamic Programming:
  - 1D (Fibonacci, Climbing Stairs)
  - 2D (Grid Paths, Minimum Path Sum)
  - Knapsack, LCS, Edit Distance

Tip: Solve 2 questions daily.

Don’t jump into hard problems right away master the patterns first.

4. HR Interview Round

This is where they judge you — your mindset, clarity, and confidence.

Your answers should feel natural, not memorized.

Resource: IndiaBix HR Interview Questions

Most Common Questions:

1. Tell me about yourself.

2. Why should we hire you?
3. What are your strengths and weaknesses?
4. Where do you see yourself in 5 years?
5. Tell me about a challenging situation you faced.
6. Why do you want to join this company?
7. Explain your final year project or internship.
8. What are your salary expectations?
9. Do you have plans for higher studies?
10. Do you prefer working in a team or individually?